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CSE 344 – Homework #4 Report

Multi-Process & Multi-Threaded Concurrent Log Analysis Pipeline

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> This report is written to satisfy Sec.17 of the homework specification. It is produced as Markdown for ease of editing; convert to `220104004007_report.pdf` using any Markdown-to-PDF pipeline (e.g. `pandoc report.md -o 220104004007_report.pdf` or VS Code "Markdown PDF" extension).

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1. Introduction

The goal of this project is to build a POSIX C pipeline that ingests multiple plain-text log files concurrently, scores each log entry against a keyword set weighted by severity level, separates priority sources into a second pipeline, and emits two synchronized output files (a human-readable text report and a compact binary checkpoint). The system is a combination of:

- **Multiple processes**: one Parent / Coordinator, N Readers (one per log file), one Dispatcher, four Analyzers (one per severity), one Aggregator.
- **Multiple threads inside each process**: Reader uses T reader threads plus one parser thread; every Analyzer uses W worker threads with `pthread_key_t` thread-local storage and a `pthread_barrier_t`; the Parent additionally runs a Watchdog thread.
- **Four shared memory regions** (A, B x 4, C, D) created via a single `mmap(MAP_SHARED | MAP_ANONYMOUS)` call with every primitive initialized with `PTHREAD_PROCESS_SHARED`.

This design isolates failure domains at the process boundary while still exploiting the low-overhead, fine-grained synchronization available to threads inside a single process.

---

2. System Architecture

```
flowchart LR
    Parent[Parent / Coordinator] -->|fork| R0[Reader 0]
    Parent -->|fork| R1[Reader 1]
    Parent -->|fork| RN[Reader N]
    Parent -->|fork| Disp[Dispatcher]
    Parent -->|fork| AE[Analyzer ERROR]
    Parent -->|fork| AW[Analyzer WARN]
    Parent -->|fork| AI[Analyzer INFO]
    Parent -->|fork| AD[Analyzer DEBUG]
    Parent -->|fork| Agg[Aggregator]
```

```
Parent -->|pthread_create| WD[Watchdog Thread]

R0 --> RegA[(Region A: Input Queue)]
R1 --> RegA
RN --> RegA
RegA -->|region_a_pop_timed| Disp
Disp -->|route by level| RegB[(Region B x 4 per-level)]
Disp -->|copy if priority| RegD[(Region D: High-Priority)]
RegB --> AE
RegB --> AW
RegB --> AI
RegB --> AD
AE -->|sem_post| RegC[(Region C: Results + 4 sems)]
AW --> RegC
AI --> RegC
AD --> RegC
RegC -->|sem_wait + cond_timedwait| Agg
RegD -->|drainer thread| Agg
Agg --> FS[results.txt + results.bin]

R0 -.heartbeat.-> WD
R1 -.heartbeat.-> WD
RN -.heartbeat.-> WD
```

Rationale for the split:

- Readers are forked first so back-pressure on Region A throttles file ingestion naturally.
- Dispatcher is a *single* process because it is the one writer of four per-level ring buffers (B) plus D; this removes any multi-producer coordination from B/D and keeps ordering deterministic.
- Each Analyzer is a *separate* process because the severity partition gives perfect isolation; a buggy regex in one level cannot crash the others.
- Aggregator is its own process so it can exit cleanly even if Parent received `SIGINT` and is tearing down.

---

3. Shared Memory Layout

All four regions live inside **one** anonymous `mmap` block created fork-before. Offsets are 64-byte aligned (cache-line-friendly) and the parent records pointers into the block inside `shm_bundle_t`.

Region	Struct type	Primitives inside	Producer(s)	Consumer(s)
A	`region_a_t`	`input_mutex`, `not_full_a`, `not_empty_a`, per-level EOF counters	All Reader parser threads	Dispatcher
B x 4	`region_b_level_t`	`level_mutex`, `not_full_b`, `not_empty_b`, `eof_posted`	Dispatcher (only)	That level's W worker threads
C	`region_c_t`	4 x `sem_t` (pshared=1), `agg_mutex`, `agg_cond`, `results_ready_mask`	Each Analyzer's reporting worker	Aggregator
D	`region_d_t`	`priority_mutex`, `not_full_d`, `not_empty_d`, `dispatcher_done`	Dispatcher	Aggregator's high-priority drainer thread

Every mutex and condvar is initialized with `pthread_mutexattr_setpshared(PTHREAD_PROCESS_SHARED)` and `pthread_condattr_setpshared(PTHREAD_PROCESS_SHARED)`. Semaphores use `sem_init(&s, 1, 0)`.

`log_entry_t` is **fixed size** (`timestamp[20] + level_str[8] + int + source[64] + message[512] + int`) so every ring buffer is a `log_entry_t buf[capacity]` stored contiguously after the header of its region.

`level_result_t` is also fixed-size (`per_keyword_score[MAX_KEYWORDS=8]`, `per_thread_score[MAX_WORKERS=64]`, `top_source[3][64]`), which makes Region C's layout *and* the binary checkpoint format entirely deterministic – the grading script can read the binary with a static struct definition.

3.1 Single `mmap` layout diagram

```
+---------- shm base ----------+
| region_a_t header            |
| log_entry_t buf[cap_A]       |
+------------------------------+
| region_b_level_t[0] header   |
| log_entry_t buf[cap_B]       |
+------------------------------+
| region_b_level_t[1] header   | (WARN)
| log_entry_t buf[cap_B]       |
+------------------------------+
| region_b_level_t[2] header   | (INFO)
| log_entry_t buf[cap_B]       |
+------------------------------+
| region_b_level_t[3] header   | (DEBUG)
| log_entry_t buf[cap_B]       |
+------------------------------+
| region_c_t (fixed size)      |
+------------------------------+
| region_d_t header            |
| log_entry_t buf[cap_D]       |
+------------------------------+
```

---

4. Lock Ordering

To keep the lock acquisition graph acyclic we enforce:

```
A   ->   B[i]   ->   C   ->   D
```

Every piece of code may hold **at most one** mutex at a time. The ordering is never violated because the data flow matches the ordering: Reader -> Dispatcher uses only A; Dispatcher -> Analyzer uses only B[i]; Analyzer -> Aggregator uses only C; Aggregator reads D independently.

Violations we actively prevent:

- Region-A push never touches B or D (Reader's parser thread). If B[i] becomes full Dispatcher waits on B's condvar *while holding B's mutex only*, never A's.
- Inside Analyzer workers TLS is thread-local so no lock is required for per-keyword accumulation; the only inter-thread synchronization is `pthread_barrier_wait` plus the process-local `agg->mtx` used by the TLS destructor and the reporting worker.
- Aggregator's high-priority drainer thread uses D's mutex only; the main Aggregator thread uses C's mutex only. There is no overlap.

Argument for freedom from deadlock: The relation "process P holds lock X while waiting for lock Y" forms a strict DAG over {A, B[0..3], C, D}. A DAG has no cycle, therefore the classical Coffman "circular wait" condition cannot hold and deadlock is impossible.

---

5. Thread Lifecycle

5.1 Reader threads

```
+-------------------------------+
| pthread_create T reader thrds |
|   each: pread byte range,     |
|   parse lines, push to        |
|   internal buffer (cond_t)    |
+-------------------------------+
| pthread_create 1 parser thr   |
|   pops internal buffer,       |
|   push to Region A,           |
```

```
|   write heartbeat every 50    |
+-------------------------------+
| pthread_join T reader threads |
| ibuf_close()                  |
| pthread_join parser           |
|   (parser then pushes 4 EOF   |
|    markers and exits)         |
+-------------------------------+
```

5.2 Analyzer worker threads

```
start -> register TID in tid_table -> loop {
    pop entry from B[level];
    if EOF: break;
    count overlapping keywords;
    accumulate into TLS per_keyword, total_weighted, per-source table;
}
pthread_barrier_wait();                // sync all W workers
if (gettid() == min_tid) {
    // reporting path
    wait until (flushed_count == W-1)  // cond_wait, no busy loop
    manually flush own TLS;
    pthread_setspecific(key, NULL);    // disable auto destructor
    compute top-3 and write Region C;
    sem_post(C.sems[level]);
    return;
} else {
    return;                            // destructor runs -> flushes TLS
}
```

5.3 Watchdog thread (parent)

`select()` over N reader pipe read-ends with 3-second timeout, parses `"[Ri] N lines processed"` heartbeats, prints a single-line snapshot each period. Exits when parent sets `shutdown_flag = 1`.

---

6. TLS Design – Why Destructors Instead of "Flush at Barrier"

A naive design would flush per-worker TLS values into Region C *right after the barrier*, inside the same worker function. We rejected that for two reasons:

1. **The spec explicitly requires a `pthread_key_t` destructor.** If we flush and zero out TLS at barrier time, `pthread_exit` afterwards has nothing to do and the destructor is effectively unused.
2. **Destructors model "definitive end of life".** Before exit the thread may still allocate, log, or change its state; the destructor is the only hook guaranteed to run with the TLS pointer still valid *and* no further user code after it. Flushing there is the textbook "resource-freeing-on-exit" pattern.

The reporting thread cannot simply use the destructor because it must *observe the flushed aggregate* before calling `sem_post` – and if it exits first, Region C is never populated. Our solution:

- All non-reporting workers return normally -> their destructors flush -> `flushed_count++` under a process-local mutex and broadcast a condvar.
- The reporting worker waits on that condvar until `flushed_count == W-1`, then performs the same flush logic manually on its own TLS array, calls `pthread_setspecific(key, NULL)` to prevent a second destructor call, writes Region C, and posts the semaphore.

This guarantees:

- Exactly `W` flushes happen (no double, no missed).
- Reporting thread's final Region C view is complete.
- TLS memory is `free()`-d exactly once per worker.

Mapping to the instructor's official Approach A / Approach B

The instructor's clarification email accepts two designs for the "barrier -> flush -> report" handoff:

- **Approach A** – explicit flush at barrier with a shared `flush_count` counter; the reporting thread (lowest TID) waits until `flush_count == W` before reading Region C.
- **Approach B** – every worker calls `pthread_exit`, the destructor flushes, and the *Analyzer main thread* `pthread_joins` all W workers before reading Region C.

Our implementation is a **hybrid that satisfies the same invariant as Approach A**:

- The lowest-TID worker plays the reporter role (as in A).
- Non-reporter workers return normally so the **TLS destructor** does the flush – closer in spirit to B, but without requiring the Analyzer main thread to perform the read.
- The reporter waits on `pthread_cond_wait` until `flushed_count == W-1`, performs its own flush (bringing the total to `W`), and only then writes Region C and signals the Aggregator.

The key invariant – *Region C is never read for final summarisation until all W workers have written their per-thread scores* – is therefore enforced by an explicit counter check, exactly as Approach A prescribes.

---

7. Dispatcher Logic – EOF Markers across Multiple Readers x Multiple Levels

Reader parser threads push **4 EOF markers per file** (one per severity level) at the end of their work. Each marker has `is_eof=1` and a valid `level_idx`. With 3 Reader processes the Dispatcher receives 12 EOF markers total (4 x 3).

```
per-level EOF counter in Region A:  eof_count_per_level[4]
on every EOF marker popped:         eof_count_per_level[lv]++

when eof_count_per_level[lv] == total_readers:
    push a single EOF marker to Region B[lv]
    set B[lv].eof_posted = 1
    broadcast B[lv].not_empty_b

when pthread_cond_timedwait on Region A times out:
    if all four eof_count_per_level == total_readers
       and Region A is empty:
        break the Dispatcher loop
        set Region D.dispatcher_done = 1
        broadcast Region D.not_empty_d
```

This approach is **N-Reader-scalable**: the counters are absolute, we do not need to know *which* Reader's EOF we've seen – only the count.

---

8. Termination Chain

8.1 Normal shutdown

```
Readers finish range -> push 4 EOFs each -> exit
Dispatcher sees enough EOFs -> forwards EOFs to all 4 B's -> sets D.done -> exit
Each Analyzer's B loop sees eof_posted + empty -> all workers barrier ->
    reporter flushes C and sem_posts -> exit
Aggregator sem_waits 4 times + timedwait on agg_cond -> drains D ->
    writes text + binary (tmp + atomic rename) -> exit
Parent waitpid loop completes -> stops Watchdog -> final summary -> munmap -> exit 0
```

8.2 SIGINT (Ctrl+C)

```
SIGINT handler (async-safe):
    write(STDERR, "[SIGINT] cleanup in progress...", ...)
    g_sigint_received = 1       // volatile sig_atomic_t
```

```
Parent main loop sees the flag:
    kill(child, SIGTERM) for every live child
    waitpid loop with WNOHANG and a 5s deadline
    any child still alive after deadline -> kill(SIGKILL)
    final waitpid to collect the last few
    set watchdog_shutdown = 1, pthread_join(watchdog)
    shm_bundle_destroy, free CLI state
    exit(1)
```

No `printf/malloc/free` runs inside the handler – only `write(2)` and a single atomic store.

---

9. Output Formats

9.1 Text (`-o`)

```
KEYWORD_LIST: error,fail,timeout
FILES: 3
TOTAL_WEIGHTED_SCORE: 114.0
HIGH_PRIORITY_SCORE: 84.0
# Levels sorted by total_weighted_score DESC
LEVEL  ENTRIES  WEIGHTED_SCORE  error   fail  timeout
ERROR       14            84.0   32.0   32.0     20.0
 WARN       12            30.0    0.0   27.0      3.0
 INFO       19             0.0    0.0    0.0      0.0
DEBUG       14             0.0    0.0    0.0      0.0
# Top-3 sources per level
ERROR kernel:32 auth:28 nginx:24
WARN auth:21 nginx:6 kernel:3
...
```

Columns are right-aligned, **exactly two spaces** between columns, one decimal for every float (`%.1f`).

9.2 Binary Checkpoint (`-O`)

`magic = 0xC5E3440B`, `version = 1`. Header followed by 4 `level_result_t` records (ERROR, WARN, INFO, DEBUG in that order).

- Write path: `fopen(".tmp")` -> `fwrite(header)` -> 4x `fwrite(result)` -> `fflush` -> `fsync` -> `fclose` -> `rename(tmp, final)`.
- Any failed partial write causes `unlink(tmp)` and a non-zero exit.

This sequence is **atomic** on the same filesystem: readers either see the old file (or no file) or the complete new file, never a truncated one.

---

10. Test Scenarios

The `tests/` directory contains fixtures and a helper `run_tests.sh` that executes every mandatory scenario from Sec.15 of the spec. Please attach uncut terminal screenshots here.

Test 1 – Baseline (3 logs, 2+2 threads, `-k error`)

> **Screenshot placeholder:** run with `-c tests/basic.conf -f tests/filter.txt -k error -t 2 -w 2 -a 16 -b 16 -d 8 -o out.txt -O out.bin` and paste the terminal + `out.txt` contents here.

Expected: binary magic starts with `0B 44 E3 C5` (little-endian of `0xC5E3440B`), totals match those printed in the final summary.

Test 2 – Multi-keyword (`-k error,fail,timeout`)

> **Screenshot placeholder.** Compare `TOTAL_WEIGHTED_SCORE` and per-level columns to a manual tally.

Test 3 – Single-threaded (`-t 1 -w 1 -a 4 -b 4 -d 2`)

Screenshot placeholder. Verifies the pipeline still closes cleanly at the minimum allowed sizes.

Test 4 – High concurrency (500-line logs, 3 consecutive runs)

Screenshot placeholder. After `bash tests/big_gen.sh 500`, run three times and paste `diff` output – results must be byte-identical.

Test 5 – Empty filter

Screenshot placeholder. `HIGH_PRIORITY_SCORE` must print `0.0` and Region D must stay empty (no priority routing lines in stdout).

Test 6 – Valgrind leak check

Result on our machine (3 Readers x 2 threads, `-k error,fail -T 5`):

```
definitely lost: 0 bytes in 0 blocks
indirectly lost: 0 bytes in 0 blocks
possibly lost:   0 bytes in 0 blocks
ERROR SUMMARY:   0 errors
```

Screenshot placeholder. Paste the full `==PID== ERROR SUMMARY` block from your own run.

Test 7 – SIGINT handling

Run Test 4 in the background, `Ctrl+C` after ~3 s. Requirements:

- Stderr prints `[SIGINT] cleanup in progress...`
- `ps -o stat,cmd | grep Z` shows no defunct children.
- `ipcs -m` does not leak anonymous shm segments (there shouldn't be any since we use `MAP_ANONYMOUS` – no System-V IPC).

Screenshot placeholder.

Test 1 - Baseline:

```
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/2201040040007$ ./analyzer -c tests/basic.conf -f tests/filter.txt -k "error" -t 2 -w 2 -a 16 -b 16 -d 8 -o out_s1.tx
t -O out_s1.bin
head -12 out_s1.txt
xxd -l 4 out_s1.bin
[PID:7711] Parent started. Files: 3, Keywords: error
[PID:7711] Shared memory initialized (A:16 B:16x4 D:8).
[PID:7711] Forking Reader 0 -> tests/logs/kernel.log
[PID:7711] Forking Reader 1 -> tests/logs/nginx.log
[PID:7712] Reader 0 started. File: tests/logs/kernel.log, Threads: 2
[PID:7711] Forking Reader 2 -> tests/logs/auth.log
[PID:7713] Reader 1 started. File: tests/logs/nginx.log, Threads: 2
[PID:7711] Forking Dispatcher
[PID:7714] Reader 2 started. File: tests/logs/auth.log, Threads: 2
[PID:7711] Forking Analyzer ERROR (index 0)
[PID:7715] Dispatcher started.
[PID:7711] Forking Analyzer WARN (index 1)
[PID:7716] Analyzer ERROR started. Workers: 2
[PID:7712][TID:0] Reader thread 0: range [0, 19808) bytes
[PID:7716][TID:7717] Worker 0 started.
[PID:7716][TID:7718] Worker 1 started.
[PID:7712][TID:0] Reader thread 1: range [19808, 39615) bytes
[PID:7714][TID:0] Reader thread 0: range [0, 19925) bytes
[PID:7713][TID:0] Reader thread 0: range [0, 19906) bytes
[PID:7714][TID:0] Reader thread 1: range [19925, 39849) bytes
[PID:7711] Forking Analyzer INFO (index 2)
[PID:7723] Analyzer WARN started. Workers: 2
[PID:7713][TID:0] Reader thread 1: range [19906, 39812) bytes
[PID:7723][TID:7726] Worker 0 started.
[PID:7723][TID:7729] Worker 1 started.
[PID:7711] Forking Analyzer DEBUG (index 3)
[PID:7730] Analyzer INFO started. Workers: 2
```

```
[PID:7730][TID:7732] Worker 0 started.
[PID:7730][TID:7733] Worker 1 started.
[PID:7711] Forking Aggregator
[PID:7734] Analyzer DEBUG started. Workers: 2
[PID:7734][TID:7735] Worker 0 started.
[PID:7734][TID:7736] Worker 1 started.
[PID:7737] Aggregator started. Waiting for 4 levels...
[PID:7711] Watchdog thread started.
[PID:7712][TID:0] Reader thread 0: finished, lines_read=252, malformed=5
[PID:7713][TID:0] Reader thread 0: finished, lines_read=251, malformed=5
[PID:7714][TID:0] Reader thread 0: finished, lines_read=250, malformed=5
[PID:7712][TID:0] Reader thread 1: finished, lines_read=248, malformed=5
[PID:7714][TID:0] Reader thread 1: finished, lines_read=250, malformed=5
[PID:7713][TID:0] Reader thread 1: finished, lines_read=249, malformed=5
[PID:7712] Parser thread: dispatched E:135 W:102 I:121 D:142 -> Region A
[PID:7712] Reader 0 exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=0 Reader 2=0 children_alive=9
[PID:7714] Parser thread: dispatched E:128 W:132 I:121 D:119 -> Region A
[PID:7714] Reader 2 exiting.
[PID:7713] Parser thread: dispatched E:126 W:129 I:123 D:122 -> Region A
[PID:7713] Reader 1 exiting.
[PID:7723][TID:7729] Worker 1 done. Entries: 162, Weighted score: 186.0
[PID:7734][TID:7735] Worker 0 done. Entries: 249, Weighted score: 92.0
[PID:7716][TID:7718] Worker 1 done. Entries: 174, Weighted score: 300.0
[PID:7730][TID:7733] Worker 1 done. Entries: 231, Weighted score: 176.0
[PID:7723][TID:7726] Worker 0 done. Entries: 201, Weighted score: 216.0
[PID:7734][TID:7736] Worker 1 done. Entries: 134, Weighted score: 57.0
[PID:7716][TID:7717] Worker 0 done. Entries: 215, Weighted score: 280.0
[PID:7730][TID:7732] Worker 0 done. Entries: 134, Weighted score: 88.0
[PID:7723][TID:7726] ** Reporting thread (lowest TID). Level: WARN **
[PID:7716][TID:7717] ** Reporting thread (lowest TID). Level: ERROR **
```

```
[PID:7730][TID:7732] ** Reporting thread (lowest TID). Level: INFO **
[PID:7730][TID:7732] Total entries: 365 | Total weighted score: 264.0
[PID:7716][TID:7717] Total entries: 389 | Total weighted score: 580.0
[PID:7723][TID:7726] Total entries: 363 | Total weighted score: 402.0
[PID:7737] ERROR result received.
[PID:7734][TID:7735] ** Reporting thread (lowest TID). Level: DEBUG **
[PID:7723] Analyzer WARN exiting.
[PID:7737] WARN result received.
[PID:7734][TID:7735] Total entries: 383 | Total weighted score: 149.0
[PID:7737] INFO result received.
[PID:7737] DEBUG result received.
[PID:7737] All results received. Writing output files...
[PID:7734] Analyzer DEBUG exiting.
[PID:7730] Analyzer INFO exiting.
[PID:7716] Analyzer ERROR exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=3
[WATCHDOG] Progress at T+3s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+6s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+8s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[PID:7715] All EOF markers forwarded to Region B. Exiting. (routed=1500 priority=333 timeouts=1)
[PID:7737] Output files written: out_s1.txt, out_s1.bin
Output files written.
[PID:7737] Aggregator exiting.
[WATCHDOG] Progress at T+11s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=0
==================================================
SYSTEM SUMMARY
Keywords     : error
Log files    : 3
Total entries: 1500
Total weighted: 1395.0
High-priority: (see out_s1.txt)
```

```
ERROR: 389 entries, score: 580.0
WARN : 363 entries, score: 402.0
INFO : 365 entries, score: 264.0
DEBUG: 383 entries, score: 149.0
==================================================
Program terminated successfully.
KEYWORD_LIST: error
FILES: 3
TOTAL_WEIGHTED_SCORE: 1395.0
HIGH_PRIORITY_SCORE: 244.0
# Levels sorted by total_weighted_score DESC
LEVEL  ENTRIES  WEIGHTED_SCORE  error
ERROR      389           580.0  580.0
WARN       363           402.0  402.0
INFO       365           264.0  264.0
DEBUG      383           149.0  149.0
# Top-3 sources per level
ERROR nginx:104 disk:88 cron:84
00000000: 0b44 e3c5                                .D..
```

Test 2 - Multi-keyword:

```
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$ ./analyzer -c tests/basic.conf -f tests/filter.txt -k "error,fail,timeout" -t 2 -w 2 -a 16 -b 16 -d 8
 -o out_s2.txt -O out_s2.bin
cat out_s2.txt
[PID:7815] Parent started. Files: 3, Keywords: error,fail,timeout
[PID:7815] Shared memory initialized (A:16 B:16x4 D:8).
[PID:7815] Forking Reader 0 -> tests/logs/kernel.log
[PID:7815] Forking Reader 1 -> tests/logs/nginx.log
[PID:7815] Forking Reader 2 -> tests/logs/auth.log
[PID:7817] Reader 1 started. File: tests/logs/nginx.log, Threads: 2
[PID:7815] Forking Dispatcher
[PID:7818] Reader 2 started. File: tests/logs/auth.log, Threads: 2
[PID:7816] Reader 0 started. File: tests/logs/kernel.log, Threads: 2
[PID:7815] Forking Analyzer ERROR (index 0)
[PID:7819] Dispatcher started.
[PID:7815] Forking Analyzer WARN (index 1)
[PID:7820] Analyzer ERROR started. Workers: 2
[PID:7815] Forking Analyzer INFO (index 2)
[PID:7821] Analyzer WARN started. Workers: 2
[PID:7820][TID:7822] Worker 0 started.
[PID:7815] Forking Analyzer DEBUG (index 3)
[PID:7820][TID:7823] Worker 1 started.
[PID:7824] Analyzer INFO started. Workers: 2
[PID:7821][TID:7825] Worker 0 started.
[PID:7821][TID:7826] Worker 1 started.
[PID:7815] Forking Aggregator
[PID:7824][TID:7828] Worker 0 started.
[PID:7827] Analyzer DEBUG started. Workers: 2
[PID:7824][TID:7829] Worker 1 started.
[PID:7827][TID:7831] Worker 0 started.
[PID:7830] Aggregator started. Waiting for 4 levels...
[PID:7815] Watchdog thread started.
[PID:7827][TID:7833] Worker 1 started.
[PID:7817][TID:0] Reader thread 0: range [0, 19906) bytes
[PID:7816][TID:0] Reader thread 0: range [0, 19808) bytes
[PID:7818][TID:0] Reader thread 0: range [0, 19925) bytes
[PID:7817][TID:0] Reader thread 1: range [19906, 39812) bytes
[PID:7818][TID:0] Reader thread 1: range [19925, 39849) bytes
[PID:7816][TID:0] Reader thread 1: range [19808, 39615) bytes
[PID:7817][TID:0] Reader thread 0: finished, lines_read=251, malformed=5
```

```
[PID:7818][TID:0] Reader thread 0: finished, lines_read=250, malformed=5
[PID:7816][TID:0] Reader thread 0: finished, lines_read=252, malformed=5
[PID:7817][TID:0] Reader thread 1: finished, lines_read=249, malformed=5
[PID:7818][TID:0] Reader thread 1: finished, lines_read=250, malformed=5
[PID:7816][TID:0] Reader thread 1: finished, lines_read=248, malformed=5
[PID:7817] Parser thread: dispatched E:126 W:129 I:123 D:122 -> Region A
[PID:7817] Reader 1 exiting.
[WATCHDOG] Progress at T+0s: Reader 0=0 Reader 1=500 Reader 2=0 children_alive=9
[PID:7816] Parser thread: dispatched E:135 W:102 I:121 D:142 -> Region A
[PID:7816] Reader 0 exiting.
[PID:7818] Parser thread: dispatched E:128 W:132 I:121 D:119 -> Region A
[PID:7818] Reader 2 exiting.
[PID:7827][TID:7833] Worker 1 done. Entries: 242, Weighted score: 170.0
[PID:7821][TID:7825] Worker 0 done. Entries: 142, Weighted score: 249.0
[PID:7820][TID:7823] Worker 1 done. Entries: 158, Weighted score: 324.0
[PID:7824][TID:7828] Worker 0 done. Entries: 234, Weighted score: 306.0
[PID:7824][TID:7829] Worker 1 done. Entries: 131, Weighted score: 142.0
[PID:7821][TID:7826] Worker 1 done. Entries: 221, Weighted score: 402.0
[PID:7827][TID:7831] Worker 0 done. Entries: 141, Weighted score: 88.0
[PID:7824][TID:7828] ** Reporting thread (lowest TID). Level: INFO **
[PID:7820][TID:7822] Worker 0 done. Entries: 231, Weighted score: 632.0
[PID:7821][TID:7825] ** Reporting thread (lowest TID). Level: WARN **
[PID:7827][TID:7831] ** Reporting thread (lowest TID). Level: DEBUG **
[PID:7824][TID:7828] Total entries: 365 | Total weighted score: 448.0
[PID:7821][TID:7825] Total entries: 363 | Total weighted score: 651.0
[PID:7827][TID:7831] Total entries: 383 | Total weighted score: 258.0
[PID:7820][TID:7822] ** Reporting thread (lowest TID). Level: ERROR **
[PID:7821] Analyzer WARN exiting.
[PID:7824] Analyzer INFO exiting.
[PID:7820][TID:7822] Total entries: 389 | Total weighted score: 956.0
[PID:7830] ERROR result received.
[PID:7830] WARN result received.
[PID:7830] INFO result received.
[PID:7830] DEBUG result received.
[PID:7827] Analyzer DEBUG exiting.
[PID:7830] All results received. Writing output files...
[PID:7820] Analyzer ERROR exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=3
[WATCHDOG] Progress at T+2s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
```

```
[WATCHDOG] Progress at T+5s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+8s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[PID:7819] All EOF markers forwarded to Region B. Exiting. (routed=1500 priority=333 timeouts=1)
[PID:7830] Output files written: out_s2.txt, out_s2.bin
Output files written.
[PID:7830] Aggregator exiting.
[WATCHDOG] Progress at T+11s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=0
==================================================
SYSTEM SUMMARY
Keywords     : error, fail, timeout
Log files    : 3
Total entries: 1500
Total weighted: 2313.0
High-priority: (see out_s2.txt)
ERROR: 389 entries, score: 956.0
WARN : 363 entries, score: 651.0
INFO : 365 entries, score: 448.0
DEBUG: 383 entries, score: 258.0
==================================================
```

```
Program terminated successfully.
KEYWORD_LIST: error,fail,timeout
FILES: 3
TOTAL_WEIGHTED_SCORE: 2313.0
HIGH_PRIORITY_SCORE: 466.0
# Levels sorted by total_weighted_score DESC
LEVEL  ENTRIES  WEIGHTED_SCORE  error   fail  timeout
ERROR      389           956.0  580.0  164.0    212.0
WARN       363           651.0  402.0  117.0    132.0
INFO       365           448.0  264.0   96.0     88.0
DEBUG      383           258.0  149.0   71.0     38.0
# Top-3 sources per level
ERROR nginx:148 cron:132 disk:132
WARN disk:108 sshd:78 cron:75
INFO sshd:70 disk:58 nginx:58
DEBUG disk:38 sshd:35 auth:33
# Per-thread contributions (weighted score)
ERROR thread_0:632.0 thread_1:324.0
WARN thread_0:249.0 thread_1:402.0
INFO thread_0:306.0 thread_1:142.0
DEBUG thread_0:88.0 thread_1:170.0
```

Test 3 - Single thread:

```
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$ ./analyzer -c tests/basic.conf -f tests/filter.txt -k "error" -t 1 -w 1 -a 4 -b 4 -d 2 -o out_s3.txt
-O out_s3.bin
grep -E '^TOTAL_WEIGHTED_SCORE|^HIGH_PRIORITY_SCORE' out_s1.txt out_s3.txt
[PID:7896] Parent started. Files: 3, Keywords: error
[PID:7896] Shared memory initialized (A:4 B:4x4 D:2).
[PID:7896] Forking Reader 0 -> tests/logs/kernel.log
[PID:7896] Forking Reader 1 -> tests/logs/nginx.log
[PID:7896] Forking Reader 2 -> tests/logs/auth.log
[PID:7897] Reader 0 started. File: tests/logs/kernel.log, Threads: 1
[PID:7898] Reader 1 started. File: tests/logs/nginx.log, Threads: 1
[PID:7896] Forking Dispatcher
[PID:7899] Reader 2 started. File: tests/logs/auth.log, Threads: 1
[PID:7896] Forking Analyzer ERROR (index 0)
[PID:7900] Dispatcher started.
[PID:7896] Forking Analyzer WARN (index 1)
[PID:7901] Analyzer ERROR started. Workers: 1
[PID:7901][TID:7902] Worker 0 started.
[PID:7896] Forking Analyzer INFO (index 2)
[PID:7903] Analyzer WARN started. Workers: 1
[PID:7903][TID:7904] Worker 0 started.
[PID:7896] Forking Analyzer DEBUG (index 3)
[PID:7905] Analyzer INFO started. Workers: 1
[PID:7897][TID:0] Reader thread 0: range [0, 39615) bytes
[PID:7905][TID:7906] Worker 0 started.
[PID:7899][TID:0] Reader thread 0: range [0, 39849) bytes
[PID:7898][TID:0] Reader thread 0: range [0, 39812) bytes
[PID:7896] Forking Aggregator
[PID:7911] Analyzer DEBUG started. Workers: 1
[PID:7911][TID:7914] Worker 0 started.
[PID:7915] Aggregator started. Waiting for 4 levels...
[PID:7896] Watchdog thread started.
[PID:7897][TID:0] Reader thread 0: finished, lines_read=500, malformed=10
[PID:7899][TID:0] Reader thread 0: finished, lines_read=500, malformed=10
[PID:7898][TID:0] Reader thread 0: finished, lines_read=500, malformed=10
[PID:7897] Parser thread: dispatched E:135 W:102 I:121 D:142 -> Region A
```

```
[PID:7897] Reader 0 exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=0 Reader 2=0[PID:7899] Parser thread: dispatched E:128 W:132 I:121 D:119 -> Region A
 children_alive=9
[PID:7899] Reader 2 exiting.
[PID:7898] Parser thread: dispatched E:126 W:129 I:123 D:122 -> Region A
[PID:7898] Reader 1 exiting.
[PID:7911][TID:7914] Worker 0 done. Entries: 383, Weighted score: 149.0
[PID:7905][TID:7906] Worker 0 done. Entries: 365, Weighted score: 264.0
[PID:7903][TID:7904] Worker 0 done. Entries: 363, Weighted score: 402.0
[PID:7901][TID:7902] Worker 0 done. Entries: 389, Weighted score: 580.0
[PID:7911][TID:7914] ** Reporting thread (lowest TID). Level: DEBUG **
[PID:7905][TID:7906] ** Reporting thread (lowest TID). Level: INFO **
[PID:7903][TID:7904] ** Reporting thread (lowest TID). Level: WARN **
[PID:7901][TID:7902] ** Reporting thread (lowest TID). Level: ERROR **
[PID:7911][TID:7914] Total entries: 383 | Total weighted score: 149.0
[PID:7905][TID:7906] Total entries: 365 | Total weighted score: 264.0
[PID:7903][TID:7904] Total entries: 363 | Total weighted score: 402.0
[PID:7901][TID:7902] Total entries: 389 | Total weighted score: 580.0
[PID:7915] ERROR result received.
[PID:7915] WARN result received.
[PID:7915] INFO result received.
[PID:7915] DEBUG result received.
[PID:7915] All results received. Writing output files...
[PID:7905] Analyzer INFO exiting.
[PID:7903] Analyzer WARN exiting.
[PID:7911] Analyzer DEBUG exiting.
[PID:7901] Analyzer ERROR exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=3
[WATCHDOG] Progress at T+3s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+6s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+9s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[PID:7900] All EOF markers forwarded to Region B. Exiting. (routed=1500 priority=333 timeouts=1)
[PID:7915] Output files written: out_s3.txt, out_s3.bin
Output files written.
[PID:7915] Aggregator exiting.
```

```
[WATCHDOG] Progress at T+12s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=0
==================================================
SYSTEM SUMMARY
Keywords     : error
Log files    : 3
Total entries: 1500
Total weighted: 1395.0
High-priority: (see out_s3.txt)
ERROR: 389 entries, score: 580.0
WARN : 363 entries, score: 402.0
INFO : 365 entries, score: 264.0
DEBUG: 383 entries, score: 149.0
==================================================
Program terminated successfully.
out_s1.txt:TOTAL_WEIGHTED_SCORE: 1395.0
out_s1.txt:HIGH_PRIORITY_SCORE: 244.0
out_s3.txt:TOTAL_WEIGHTED_SCORE: 1395.0
out_s3.txt:HIGH_PRIORITY_SCORE: 244.0
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$
```

Test 4 - High concurrency:

```
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$ for i in 1 2 3; do ./analyzer -c tests/basic.conf -f tests/filter.txt -k "error,fail,timeout" -t 8 -w
 6 -a 128 -b 128 -d 32 -o out_s4_$i.txt -O out_s4_$i.bin 2>&1 | tail -2; done
md5sum out_s4_1.txt out_s4_2.txt out_s4_3.txt
md5sum out_s4_1.bin out_s4_2.bin out_s4_3.bin
==================================================
Program terminated successfully.
==================================================
Program terminated successfully.
==================================================
Program terminated successfully.
6081e75bb7dccc078d4120d6a4594383  out_s4_1.txt
d39706893e7514cce5384e4a078c0b51  out_s4_2.txt
5ae45c0927f3160eec48f708a7dd26cd  out_s4_3.txt
49344f6ad01435d203de55a1721ef20f  out_s4_1.bin
44d36468bee78e05cf8fc18fdfe0636c  out_s4_2.bin
20e5a618701caec88306ea5e7aafaba4  out_s4_3.bin
```

Test 5 - No priority matches:

```
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$ ./analyzer -c tests/basic.conf -f tests/empty_filter.txt -k "error" -t 2 -w 2 -a 16 -b 16 -d 8 -o out
_s5.txt -O out_s5.bin
grep '^HIGH_PRIORITY_SCORE' out_s5.txt
[PID:8276] Parent started. Files: 3, Keywords: error
[PID:8276] Shared memory initialized (A:16 B:16x4 D:8).
[PID:8276] Forking Reader 0 -> tests/logs/kernel.log
[PID:8276] Forking Reader 1 -> tests/logs/nginx.log
[PID:8277] Reader 0 started. File: tests/logs/kernel.log, Threads: 2
[PID:8276] Forking Reader 2 -> tests/logs/auth.log
[PID:8278] Reader 1 started. File: tests/logs/nginx.log, Threads: 2
[PID:8276] Forking Dispatcher
[PID:8279] Reader 2 started. File: tests/logs/auth.log, Threads: 2
[PID:8276] Forking Analyzer ERROR (index 0)
[PID:8280] Dispatcher started.
[PID:8276] Forking Analyzer WARN (index 1)
[PID:8281] Analyzer ERROR started. Workers: 2
[PID:8276] Forking Analyzer INFO (index 2)
[PID:8282] Analyzer WARN started. Workers: 2
[PID:8281][TID:8283] Worker 0 started.
[PID:8281][TID:8284] Worker 1 started.
[PID:8282][TID:8285] Worker 0 started.
[PID:8282][TID:8286] Worker 1 started.
[PID:8276] Forking Analyzer DEBUG (index 3)
[PID:8287] Analyzer INFO started. Workers: 2
[PID:8276] Forking Aggregator
[PID:8287][TID:8289] Worker 0 started.
[PID:8288] Analyzer DEBUG started. Workers: 2
[PID:8287][TID:8290] Worker 1 started.
[PID:8288][TID:8291] Worker 0 started.
[PID:8292] Aggregator started. Waiting for 4 levels...
[PID:8288][TID:8293] Worker 1 started.
[PID:8276] Watchdog thread started.
[PID:8279][TID:0] Reader thread 0: range [0, 19925) bytes
[PID:8277][TID:0] Reader thread 0: range [0, 19808) bytes
[PID:8278][TID:0] Reader thread 0: range [0, 19906) bytes
[PID:8277][TID:0] Reader thread 1: range [19808, 39615) bytes
[PID:8278][TID:0] Reader thread 1: range [19906, 39812) bytes
[PID:8279][TID:0] Reader thread 1: range [19925, 39849) bytes
[PID:8277][TID:0] Reader thread 0: finished, lines_read=252, malformed=5
[PID:8279][TID:0] Reader thread 0: finished, lines_read=250, malformed=5
[PID:8278][TID:0] Reader thread 0: finished, lines_read=251, malformed=5
[PID:8279][TID:0] Reader thread 1: finished, lines_read=250, malformed=5
[PID:8277][TID:0] Reader thread 1: finished, lines_read=248, malformed=5
[PID:8278][TID:0] Reader thread 1: finished, lines_read=249, malformed=5
[PID:8279] Parser thread: dispatched E:128 W:132 I:121 D:119 -> Region A
[PID:8279] Reader 2 exiting.
[WATCHDOG] Progress at T+0s: Reader 0=0 Reader 1=0 Reader 2=500 children_alive=9
[PID:8277] Parser thread: dispatched E:135 W:102 I:121 D:142 -> Region A
[PID:8277] Reader 0 exiting.
[PID:8278] Parser thread: dispatched E:126 W:129 I:123 D:122 -> Region A
[PID:8278] Reader 1 exiting.
[PID:8281][TID:8284] Worker 1 done. Entries: 241, Weighted score: 364.0
[PID:8282][TID:8285] Worker 0 done. Entries: 201, Weighted score: 183.0
[PID:8287][TID:8289] Worker 0 done. Entries: 198, Weighted score: 166.0
[PID:8288][TID:8293] Worker 1 done. Entries: 198, Weighted score: 71.0
[PID:8282][TID:8286] Worker 1 done. Entries: 162, Weighted score: 219.0
[PID:8281][TID:8283] Worker 0 done. Entries: 148, Weighted score: 216.0
[PID:8287][TID:8290] Worker 1 done. Entries: 167, Weighted score: 98.0
[PID:8281][TID:8283] ** Reporting thread (lowest TID). Level: ERROR **
[PID:8288][TID:8291] Worker 0 done. Entries: 185, Weighted score: 78.0
[PID:8282][TID:8285] ** Reporting thread (lowest TID). Level: WARN **
[PID:8282][TID:8285] Total entries: 363 | Total weighted score: 402.0
[PID:8288][TID:8291] ** Reporting thread (lowest TID). Level: DEBUG **
[PID:8287][TID:8289] ** Reporting thread (lowest TID). Level: INFO **
[PID:8287][TID:8289] Total entries: 365 | Total weighted score: 264.0
[PID:8281][TID:8283] Total entries: 389 | Total weighted score: 580.0
[PID:8292] ERROR result received.
[PID:8292] WARN result received.
[PID:8292] INFO result received.
[PID:8282] Analyzer WARN exiting.
[PID:8287] Analyzer INFO exiting.
[PID:8281] Analyzer ERROR exiting.
[PID:8288][TID:8291] Total entries: 383 | Total weighted score: 149.0
[PID:8292] DEBUG result received.
[PID:8292] All results received. Writing output files...
[PID:8288] Analyzer DEBUG exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=3
[WATCHDOG] Progress at T+3s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+6s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+9s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[PID:8280] All EOF markers forwarded to Region B. Exiting. (routed=1500 priority=0 timeouts=1)
[PID:8292] Output files written: out_s5.txt, out_s5.bin
Output files written.
[PID:8292] Aggregator exiting.
[WATCHDOG] Progress at T+12s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=0
==================================================
SYSTEM SUMMARY
Keywords     : error
Log files    : 3
Total entries: 1500
Total weighted: 1395.0
High-priority: (see out_s5.txt)
ERROR: 389 entries, score: 580.0
WARN : 363 entries, score: 402.0
INFO : 365 entries, score: 264.0
DEBUG: 383 entries, score: 149.0
==================================================
Program terminated successfully.
HIGH_PRIORITY_SCORE: 0.0
```

Test 6 - Valgrind:

```
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/2201040040007$ valgrind --leak-check=full --error-exitcode=1 --child-silent-after-fork=yes ./analyzer -c tests/basic
.conf -f tests/filter.txt -k "error" -t 1 -w 1 -a 8 -b 8 -d 4 -o out_s6.txt -O out_s6.bin
echo "valgrind exit_code = $?"
==8354== Memcheck, a memory error detector
==8354== Copyright (C) 2002-2017, and GNU GPL'd, by Julian Seward et al.
==8354== Using Valgrind-3.18.1 and LibVEX; rerun with -h for copyright info
==8354== Command: ./analyzer -c tests/basic.conf -f tests/filter.txt -k error -t 1 -w 1 -a 8 -b 8 -d 4 -o out_s6.txt -O out_s6.bin
==8354==
[PID:8354] Parent started. Files: 3, Keywords: error
[PID:8354] Shared memory initialized (A:8 B:8x4 D:4).
[PID:8354] Forking Reader 0 -> tests/logs/kernel.log
[PID:8354] Forking Reader 1 -> tests/logs/nginx.log
[PID:8354] Forking Reader 2 -> tests/logs/auth.log
[PID:8355] Reader 0 started. File: tests/logs/kernel.log, Threads: 1
[PID:8354] Forking Dispatcher
[PID:8354] Forking Analyzer ERROR (index 0)
[PID:8356] Reader 1 started. File: tests/logs/nginx.log, Threads: 1
[PID:8354] Forking Analyzer WARN (index 1)
[PID:8357] Reader 2 started. File: tests/logs/auth.log, Threads: 1
[PID:8354] Forking Analyzer INFO (index 2)
[PID:8358] Dispatcher started.
[PID:8354] Forking Analyzer DEBUG (index 3)
[PID:8354] Forking Aggregator
[PID:8359] Analyzer ERROR started. Workers: 1
[PID:8356][TID:0] Reader thread 0: range [0, 39812) bytes
[PID:8355][TID:0] Reader thread 0: range [0, 39615) bytes
[PID:8357][TID:0] Reader thread 0: range [0, 39849) bytes
[PID:8360] Analyzer WARN started. Workers: 1
[PID:8361] Analyzer INFO started. Workers: 1
[PID:8362] Analyzer DEBUG started. Workers: 1
[PID:8363] Aggregator started. Waiting for 4 levels...
[PID:8354] Watchdog thread started.
[PID:8359][TID:8367] Worker 0 started.
[PID:8360][TID:8368] Worker 0 started.
[PID:8362][TID:8371] Worker 0 started.
[PID:8361][TID:8370] Worker 0 started.
[PID:8357][TID:0] Reader thread 0: finished, lines_read=500, malformed=10
[PID:8356][TID:0] Reader thread 0: finished, lines_read=500, malformed=10
[PID:8355][TID:0] Reader thread 0: finished, lines_read=500, malformed=10
```

```
[PID:8357] Parser thread: dispatched E:128 W:132 I:121 D:119 -> Region A
[PID:8357] Reader 2 exiting.
[WATCHDOG] Progress at T+0s: Reader 0=0 Reader 1=0 Reader 2=500 children_alive=9
[PID:8356] Parser thread: dispatched E:126 W:129 I:123 D:122 -> Region A
[PID:8356] Reader 1 exiting.
[PID:8355] Parser thread: dispatched E:135 W:102 I:121 D:142 -> Region A
[PID:8355] Reader 0 exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=9
[PID:8362][TID:8371] Worker 0 done. Entries: 383, Weighted score: 149.0
[PID:8360][TID:8368] Worker 0 done. Entries: 363, Weighted score: 402.0
[PID:8361][TID:8370] Worker 0 done. Entries: 365, Weighted score: 264.0
[PID:8362][TID:8371] ** Reporting thread (lowest TID). Level: DEBUG **
[PID:8361][TID:8370] ** Reporting thread (lowest TID). Level: INFO **
[PID:8360][TID:8368] ** Reporting thread (lowest TID). Level: WARN **
[PID:8359][TID:8367] Worker 0 done. Entries: 389, Weighted score: 580.0
[PID:8359][TID:8367] ** Reporting thread (lowest TID). Level: ERROR **
[PID:8362][TID:8371] Total entries: 383 | Total weighted score: 149.0
[PID:8360][TID:8368] Total entries: 363 | Total weighted score: 402.0
[PID:8361][TID:8370] Total entries: 365 | Total weighted score: 264.0
[PID:8359][TID:8367] Total entries: 389 | Total weighted score: 580.0
[PID:8362] Analyzer DEBUG exiting.
[PID:8363] ERROR result received.
[PID:8363] WARN result received.
[PID:8360] Analyzer WARN exiting.
[PID:8363] INFO result received.
[PID:8363] DEBUG result received.
[PID:8363] All results received. Writing output files...
[PID:8361] Analyzer INFO exiting.
[PID:8359] Analyzer ERROR exiting.
[WATCHDOG] Progress at T+3s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+6s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[WATCHDOG] Progress at T+9s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
[PID:8358] All EOF markers forwarded to Region B. Exiting. (routed=1500 priority=333 timeouts=1)
[PID:8363] Output files written: out_s6.txt, out_s6.bin
Output files written.
[PID:8363] Aggregator exiting.
[WATCHDOG] Progress at T+10s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=0
==================================================
SYSTEM SUMMARY
```

```
Keywords     : error
Log files    : 3
Total entries: 1500
Total weighted: 1395.0
High-priority: (see out_s6.txt)
ERROR: 389 entries, score: 580.0
WARN : 363 entries, score: 402.0
INFO : 365 entries, score: 264.0
DEBUG: 383 entries, score: 149.0
==================================================
Program terminated successfully.
==8354==
==8354== HEAP SUMMARY:
==8354==     in use at exit: 0 bytes in 0 blocks
==8354==   total heap usage: 31 allocs, 31 frees, 13,257 bytes allocated
==8354==
==8354== All heap blocks were freed -- no leaks are possible
==8354==
==8354== For lists of detected and suppressed errors, rerun with: -s
==8354== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
valgrind exit_code = 0
```

Test 7 – CTRL + C:

```
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$ ./analyzer -c tests/basic.conf -f tests/filter.txt -k "error" -t 2 -w 2 -a 16 -b 16 -d 8 -o out_s7.tx
t -O out_s7.bin
[PID:8445] Parent started. Files: 3, Keywords: error
[PID:8445] Shared memory initialized (A:16 B:16x4 D:8).
[PID:8445] Forking Reader 0 -> tests/logs/kernel.log
[PID:8445] Forking Reader 1 -> tests/logs/nginx.log
[PID:8445] Forking Reader 2 -> tests/logs/auth.log
[PID:8446] Reader 0 started. File: tests/logs/kernel.log, Threads: 2
[PID:8445] Forking Dispatcher
[PID:8447] Reader 1 started. File: tests/logs/nginx.log, Threads: 2
[PID:8448] Reader 2 started. File: tests/logs/auth.log, Threads: 2
[PID:8445] Forking Analyzer ERROR (index 0)
[PID:8449] Dispatcher started.
[PID:8445] Forking Analyzer WARN (index 1)
[PID:8450] Analyzer ERROR started. Workers: 2
[PID:8445] Forking Analyzer INFO (index 2)
[PID:8451] Analyzer WARN started. Workers: 2
[PID:8450][TID:8452] Worker 0 started.
[PID:8450][TID:8454] Worker 1 started.
[PID:8445] Forking Analyzer DEBUG (index 3)
[PID:8453] Analyzer INFO started. Workers: 2
[PID:8445] Forking Aggregator
[PID:8455] Analyzer DEBUG started. Workers: 2
[PID:8451][TID:8456] Worker 0 started.
[PID:8451][TID:8457] Worker 1 started.
[PID:8453][TID:8458] Worker 0 started.
[PID:8455][TID:8459] Worker 0 started.
[PID:8455][TID:8460] Worker 1 started.
[PID:8453][TID:8461] Worker 1 started.
[PID:8462] Aggregator started. Waiting for 4 levels...
[PID:8445] Watchdog thread started.
[PID:8446][TID:0] Reader thread 0: range [0, 19808) bytes
[PID:8448][TID:0] Reader thread 0: range [0, 19925) bytes
[PID:8447][TID:0] Reader thread 0: range [0, 19906) bytes
[PID:8447][TID:0] Reader thread 1: range [19906, 39812) bytes
[PID:8448][TID:0] Reader thread 1: range [19925, 39849) bytes
[PID:8446][TID:0] Reader thread 1: range [19808, 39615) bytes
[PID:8447][TID:0] Reader thread 0: finished, lines_read=251, malformed=5
[PID:8446][TID:0] Reader thread 0: finished, lines_read=252, malformed=5
```

```
[PID:8448][TID:0] Reader thread 0: finished, lines_read=250, malformed=5
[PID:8448][TID:0] Reader thread 1: finished, lines_read=250, malformed=5
[PID:8446][TID:0] Reader thread 1: finished, lines_read=248, malformed=5
[PID:8447][TID:0] Reader thread 1: finished, lines_read=249, malformed=5
[PID:8448] Parser thread: dispatched E:128 W:132 I:121 D:119 -> Region A
[PID:8448] Reader 2 exiting.
[WATCHDOG] Progress at T+0s: Reader 0=0 Reader 1=0 Reader 2=500 children_alive=9
[PID:8446] Parser thread: dispatched E:135 W:102 I:121 D:142 -> Region A
[PID:8446] Reader 0 exiting.
[PID:8447] Parser thread: dispatched E:126 W:129 I:123 D:122 -> Region A
[PID:8447] Reader 1 exiting.
[PID:8450][TID:8454] Worker 1 done. Entries: 185, Weighted score: 316.0
[PID:8451][TID:8457] Worker 1 done. Entries: 249, Weighted score: 282.0
[PID:8453][TID:8458] Worker 0 done. Entries: 92, Weighted score: 78.0
[PID:8455][TID:8460] Worker 1 done. Entries: 120, Weighted score: 42.0
[PID:8450][TID:8452] Worker 0 done. Entries: 204, Weighted score: 264.0
[PID:8451][TID:8456] Worker 0 done. Entries: 114, Weighted score: 120.0
[PID:8453][TID:8461] Worker 1 done. Entries: 273, Weighted score: 186.0
[PID:8455][TID:8459] Worker 0 done. Entries: 263, Weighted score: 107.0
[PID:8450][TID:8452] ** Reporting thread (lowest TID). Level: ERROR **
[PID:8451][TID:8456] ** Reporting thread (lowest TID). Level: WARN **
[PID:8455][TID:8459] ** Reporting thread (lowest TID). Level: DEBUG **
[PID:8453][TID:8458] ** Reporting thread (lowest TID). Level: INFO **
[PID:8451][TID:8456] Total entries: 363 | Total weighted score: 402.0
[PID:8455][TID:8459] Total entries: 383 | Total weighted score: 149.0
[PID:8453][TID:8458] Total entries: 365 | Total weighted score: 264.0
[PID:8450][TID:8452] Total entries: 389 | Total weighted score: 580.0
[PID:8462] ERROR result received.
[PID:8462] WARN result received.
[PID:8462] INFO result received.
[PID:8462] DEBUG result received.
[PID:8462] All results received. Writing output files...
[PID:8455] Analyzer DEBUG exiting.
[PID:8453] Analyzer INFO exiting.
[PID:8451] Analyzer WARN exiting.
[PID:8450] Analyzer ERROR exiting.
[WATCHDOG] Progress at T+0s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=6
[WATCHDOG] Progress at T+3s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=2
^C
```

```
[SIGINT] cleanup in progress...

[SIGINT] cleanup in progress...

[SIGINT] cleanup in progress...
[WATCHDOG] Progress at T+6s: Reader 0=500 Reader 1=500 Reader 2=500 children_alive=0
[main] shutting down with errors (sigint=1 failed=1)
sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$ ps aux | grep -E 'analyzer|defunct' | grep -v grep
ipcs -m | head -10

------ Shared Memory Segments --------
key        shmid      owner      perms      bytes      nattch     status

sc@SC-DESKOP:/mnt/c/Users/Cengiz/Desktop/220104004007$
```

11. Challenges & Solutions

1. **Dispatcher <-> Aggregator deadlock.** First implementation had the Aggregator waiting on Region C *before* consuming Region D. With small `-d`, Region D filled, the Dispatcher blocked pushing to D, Region A filled, Reader parsers blocked pushing to A – classic four-way deadlock. **Fix:** the Aggregator now spawns a dedicated `drain` thread that pops from Region D in parallel with the main thread waiting on the four semaphores. Back-pressure is eliminated without violating lock ordering (drainer only touches D's mutex).

2. **TLS-destructor vs reporting-thread race.** If the destructor fires for the reporting worker before it writes Region C, the reporting worker loses access to its TLS. **Fix:** the reporting worker drains TLS manually, then calls `pthread_setspecific(key, NULL)` to disable the auto-destructor. All other workers rely on the destructor and bump `flushed_count` under a process-local mutex; the reporting worker waits for `flushed_count == W-1` on a condvar instead of busy-waiting.

3. **Overlapping keyword substring search.** `strstr`-based scanning misses `"errorerror"` counted as two hits of `"error"`. **Fix:** a sliding window that advances one byte at a time, using `memcmp` for the tail comparison. The helper lives in `common.h` as `count_overlapping`.

4. **Race-free EOF termination across multiple readers *and* levels.** Four independent per-level counters in Region A plus a single timeout check turn "when have all readers finished pushing everything?" into a simple numeric invariant (`sum(counters) == 4 x N_readers`) that the Dispatcher can poll after every `pthread_cond_timedwait`.

5. **Silent validation of malformed lines.** The spec forbids reporting malformed lines to stderr. Our parser returns `-1` without printing and the caller just counts them (`malformed` field) so the `fprintf` shown on thread-join is purely informational and never leaks user data.

6. **Reader byte-range boundary duplication.** Each reader-process spawns `-t` worker threads that share a single fd and split the file via `pread()`. The original implementation set a `past_end` flag only at the end of each 4 KiB chunk loop, which meant that after a thread crossed its `end_off`, it could still process multiple additional lines within the same chunk before the flag was observed. This double-counted lines that straddled a boundary when `end_off` fell near the middle of a chunk. **Fix:** evaluate `(pos + i) >= end_off` the instant a `\n` is consumed and return immediately. Now counts match exactly for `-t 1`, `-t 2`, `-t 4`, and `-t 8` on the same input, which is verified in the Test-3 vs Test-1 equality check.

7. **Deterministic top-3 sources under ties.** Selection-sort over `src_tbl` is not stable across runs when multiple sources share the same hit count, because `src_tbl` insertion order depends on which worker happened to encounter that source first. **Fix:** add a secondary ascending `strcmp` key when hit counts are equal – so a tie between (`auth:116`, `cache:116`) always prints `auth` first. This closes a small determinism gap surfaced by Test 4.

8. **Parent hang in `shm_bundle_destroy` during forced shutdown.** When SIGINT causes the parent to `SIGKILL` stuck children, those children may have been holding a `PTHREAD_PROCESS_SHARED` mutex or sleeping on a cond. Calling `pthread_mutex_destroy`/`pthread_cond_destroy`/ `sem_destroy` from the parent could then wedge on the corresponding internal futex. **Fix:** since every primitive lives inside the monolithic anonymous `mmap` region, the `munmap` alone reclaims the kernel-side state cleanly. The destroy calls were removed, and the parent now exits promptly even in the worst-case SIGKILL path.

9. **Per-thread contribution lines are intentionally non-invariant.** The four `# Per-thread contributions (weighted score)` lines reflect the actual work done by each worker thread, which depends on how the OS scheduler and mutex fairness interleaved pops from Region B. They are included for transparency and per-thread accountability, not as a determinism invariant. The invariant outputs – `TOTAL_WEIGHTED_SCORE`, `HIGH_PRIORITY_SCORE`, per-level sorted levels, per-level entry counts, per-keyword scores, top-3 sources – are byte-identical across runs of Test 4.

---

12. Build & Run

```
make                                        # release
make debug                                  # -O0 -g
make tsan                                   # ThreadSanitizer
make clean                                  # remove binaries and .d files

./analyzer -c tests/basic.conf \
```

```
            -f tests/filter.txt \
            -k "error,fail,timeout" \
            -t 2 -w 2 -a 16 -b 16 -d 8 -T 5 \
            -o out.txt -O out.bin
```

`make` is deterministic; no target auto-runs the binary.

---

13. References

- POSIX Threads specification (IEEE 1003.1-2017): `pthread_cond_timedwait`, `pthread_key_create`, `pthread_barrier_wait`, `pthread_mutexattr_setpshared`.
- Linux manual pages: `mmap(2)`, `sem_init(3)`, `syscall(2)` for `SYS_gettid`, `rename(2)` for atomicity guarantees.
- Course notes on concurrent programming and synchronization (CSE 344 Spring 2026 lectures).